

206.1 - Project

- ▶ Data analysis and storytelling
- ▶ 8 groups (2 to 3 students per group)
- ▶ 5 topics
 - 1. Greenhouse gases
 - 2. Vehicle Safety
 - 3. Global Food Security Intelligence
 - 4. Energy efficient cars
 - 5. NBA stars
- ▶ For each topic:
 - One group tells a story in favor of A,
 - The other group in favor of B

206.1 - Project

- ▶ Topic 1: Greenhouse gases
 - Group A: “Efforts should be made in the transport domain to reduce emissions”
 - Group B: “Efforts should be made in other domains for greater impact”
 - Data
 - Starting point:
 - <https://opendata.swiss/fr/dataset/treibhausgasemissionen-nach-ve-rursachergruppen5>
 - Other possible sources: <https://opendata.swiss/fr/dataset/raum-und-umwelt-im-europaischen-vergleich3>
<https://opendata.swiss/fr/dataset/luftemissionskonten-der-haushalte-und-der-wirtschaft-nach-branchen-aggregiert-nach-abs-chnitten4>
<https://opendata.swiss/fr/dataset/energieeinsatzkonto-der-haushalte-und-der-wirtschaft-nach-branchen-und-energieträgern-in-tj4>
<https://opendata.swiss/fr/dataset/tabellen-zu-den-ausgleichspositionen-zwischen-dem-treibhausgas-und-luftschadstoff-inve-ntar-und-4>
 - Can go look for more data on opendata.swiss and other (trustable) sources

206.1 - Project

- ▶ Topic 2a: Vehicle Safety Is Improving
 - A: “Modern vehicles are safer than ever.”
 - B: “Modern vehicles are becoming more failure-prone and over-engineered.”
 - Data
 - <https://www.kaggle.com/datasets/jahnavikachhia23/nhtsa-vehicle-safety-recall-intelligence-20102026>
 - Can go look for more data from other trustable sources

206.1 - Project

- ▶ Topic 2b: The Takata Crisis Changed Automotive Safety Culture
 - A: “Post-Takata, manufacturers became more proactive and transparent.”
 - B: “Recall growth after Takata reflects continued systemic problems.”
 - Data
 - <https://www.kaggle.com/datasets/jahnavikachhia23/nhtsa-vehicle-safety-recall-intelligence-20102026>
 - Can go look for more data from other trustable sources

206.1 - Project

- ▶ Topic 3: Global Food Security Intelligence
 - Group A: “Economic growth is the primary solution to hunger”
 - Group B: “Economic growth alone does not solve food insecurity”
 - Data
 - <https://www.kaggle.com/datasets/kanchana1990/global-food-security-intelligence>
 - Can go look for more data from other trustable sources

206.1 - Project

- ▶ Topic 4: Energy efficient cars
 - Group A: “Car engines constantly improve efficiency, so transport becomes less impactful”
 - Group B: “Engine improvement has no actual impact, as cars become larger/heavier and thus actual efficiency is virtually constant”
 - Data
 - <https://www.kaggle.com/datasets/jahaidulislam/car-specification-dataset-1945-2020?select=Car+Dataset+1945-2020.csv>
 - Can go look for more data from other trustable sources

206.1 - Project

- ▶ Topic 5: NBA stars
 - Group A: “Star players determine game outcomes”
 - Group B: “Teams win games, not stars”
 - Data
 - <https://www.kaggle.com/datasets/chevronronson/nba-stats-dataset>
 - Can go look for more data from other trustable sources

206.1 - Project - Evaluation

- ▶ Continuous evaluation: 5 milestones monitored every week
 - Week 1: Gather data, load and visualize it, clean data if needed
 - Week 2: First draft of narrative (can be changed later), make plots, support the narrative
 - Week 3: Refine narrative with obtained evidence, iterate, make story convincing
 - Week 4: Introduce a lie, finalize report and presentation
 - Week 5: Storytelling Presentation
- ▶ Report (in english), 2 parts:
 - Part 1: This is our claim, this supports our claim (storytelling with the main results)
 - Part 2: How did we come to the results and story (data source, cleaning, challenges, analyses you dropped, etc.). Mention the lie at the end, separated from the main part so we don't see it before the presentation.
- ▶ Oral presentation
 - 10 min presentation with slides + Questions (additional to the 10 min)
 - **Data storytelling** (effectively tell a story with data)
 - Visualizations
 - Narrative

206.1 - Project - Evaluation

Data Collection and Processing	
Criteria	Description
Data Collection	Quality and relevance of collected data
Data Cleaning	Thoroughness and effectiveness of data cleaning processes
Data Analysis	Appropriateness and depth of data analysis methods
Data Integration	Effective integration of different data sources
Visualization and Storytelling	
Criteria	Description
Visualization Quality	Clarity, accuracy, and effectiveness of visualizations
Creativity	Innovation and creativity in data presentation
Storytelling	Coherence and engagement in presenting the data story
Report Evaluation	
Criteria	Description
Introduction	Clear explanation of the project's goals and objectives
Methodology	Detailed and appropriate description of the methods used
Results	Presentation and interpretation of results, including insights from visualizations
Discussion	Critical analysis of results, including limitations and implications
Conclusion	Summary of findings and their relevance
Presentation Evaluation	
Criteria	Description
Clarity	Clear explanation of project, methods, and results
Engagement	Ability to engage and maintain audience interest
Visual Aids	Effective use of slides and other visual aids to enhance understanding
Timing	Adherence to time limits
Response to Questions	Ability to answer questions clearly and accurately
Continuous Evaluation	
Criteria	Description
Progress Updates	Regular and thorough updates on project progress
Meeting Deadlines	Consistently meeting project milestones and deadlines
Collaboration	Effective collaboration with team members and/or supervisors
Adaptability	Ability to adapt to feedback and make necessary adjustments

206.1 - Project

- ▶ Introduce a “lie” in the presentation
 - The audience (profs and other students) must find it
 - Examples:
 - Choice of scale in graphs
 - Generalising from very small samples / non-representative sample (Biased sample selection)
 - Selective data points / time periods (picking the past 6 months because it shows a trend that supports our claim)
 - Misleading axis labels (y-axis does not start at 0)
 - **Not modifying/fabricating the data**, the lie is rather in the conclusion drawn
 - Should be somehow possible to spot by the audience (not too obvious, not impossible to find)